FEDERAL SPECIFICATION

ALUM, AMMONIA (ALUMINUM AMMONIUM SULFATE), TECHNICAL GRADE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers ammonia alum (aluminum ammonium sulfate), $Al_2(SO_4)_3(NH_4)_2SO_4 \cdot 24\ H_2O$, in nut size for use as a coagulant in water purification units and in capsule for use in water testing.

1.2 Classification. — Ammonia alum conforming to the requirements of this specification shall be of one type and of the following classes, as specified (see 6.1):

Class 1.—Nut size
Class 2.—Capsule

2. APPLICABLE SPECIFICATIONS, STANDARD, AND OTHER PUBLICATION

2.1 The following specifications and standard of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

P–S–536—Soap and Soap-Products (Including Synthetic Detergents); Methods of Sampling and Testing.
NN–B–591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).
RR–S–366—Sieves; Standard, Testing.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

Military Specifications:

JAN–P–103—Packaging and Packing for Overseas Shipment—Boxes; Wood-Cleated, Solid-Fiberboard.
JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

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JAN-D-111—Drums, Fiberboard (Overseas Type).

JAN-P-112—Packaging and Packing for Overseas Shipment—Drums, Plywood.

MIL-C-124—Containers (Can, Pails, and Drums) Metal (for Other than Subsistence Items).

JAN-P-133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

JAN-P-138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.

MIL-A-140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

MIL-L-10547—Liners, Case, Waterproof.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of Military specifications and standard references above, required by the contractor in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publication.—The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification.

American Society for Testing Materials Publication:

E-11-39—Sieves for Testing Purposes.

(Copies of the American Society for Testing Materials publication may be obtained from the Society, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Chemical requirements. — The ammonia alum shall conform to the chamical requirements shown in table I.

TABLE I.—*Chemical requirements*

	Percent by weight
Alumina (Al_2O_3)—minimum	11.00
Ammonia (NH_3)—minimum	3.50
Sulfur trioxide (SO_3)—minimum	35.00
Water insoluble matter—maximum	0.005
Moisture—maximum	0.20
Other impurities	See 3.2
Water of hydration	Remainder

3.2 Impurities.—The alum shall contain no impurity which will produce an unpleasant taste or odor in chlorinated or unchlorinated water. The alum shall contain no soluble material in quantities capable of producing deleterious or injurious effects to the health of those consuming the water which has been properly treated with the alum (see 6.2).

3.3 Storage life. — The ammonia alum shall not decompose, powder, or break down into fine during 6 months storage (see 6.2).

3.4 Class 1.—Class 1 alum, before shipment, shall pass 100 percent through a 2-inch sieve conforming to A.S.T.M. Designation E-11-39 and not less than 96.8 percent shall be retained on a No. 6 standard sieve conforming to Federal Specification RR-S-366 when tested as specified in 4.3.6.

3.5 Class 2.—Class 2 alum shall be furnished in an easily opened gelatin or plastic capsule containing 0.99 gm. plus or minus 0.05 gm. of ammonia alum. When necessary, to secure proper weights in machine filling of the capsules, only sodium chloride shall be used.

3.6 Workmanship.—Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of commodity.

[1] Applicable to Navy purchases only.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—For purposes of sampling, a lot shall consist of all material offered for delivery on the same day.

4.2 **Sampling.** — Ten containers shall be taken at random from each lot. A representative sample shall be taken from each container and the samples divided into 2 groups of 5 specimens each. A composite sample shall be prepared from each group. (The capsules shall be opened and the contents removed and thoroughly mixed.) Each composite sample shall be not less than 3 pounds nor more than 10 pounds for class 1, and not less than 90 capsules for class 2. If there are fewer than 10 containers in the lot, a representative sample shall be taken from each container, and divided into 2 approximately equal parts. Each composite sample shall be placed in a clean dry container, sealed, and labeled in a manner that will identify the specimen with the containers and lot represented.

4.3 **Inspection and tests.**—The alum and the preparation for delivery thereof shall be given such inspection and tests as may be necessary to determine conformance with this specification. (For Naval purchases the general inspection procedure shall be in accordance with the General Specifications for Inspection of Material.)

4.3.1 *Preparation of solution.*—Add 5 gm. plus or minus 0.01 gm. of the sample to 100 ml. of distilled water, digest on a steam bath for 5 minutes. Filter into a 500 ml. volumetric flask, wash the insoluble residue on the filter with distilled water. Cool the contents of the flask to room temperature (20° to 30° C.; 68° to 86° F.), add 2 ml. of hydrochloric acid (concentrated), and dilute to exactly 500 ml. Retain the solution for analyses. Reserve the insoluble residue for 4.3.2.

4.3.2 *Water insoluble matter.*—Place the insoluble residue and filter paper (see 4.3.1) in a tared crucible, dry at 100° to 105° C. and ignite to constant weight.

$$\text{Percent insoluble matter} = \frac{100A}{W}$$

where:

A = weight of insoluble residue
W = weight of sample

4.3.3 *Aluminum oxide content.* — Take a 50 ml. aliquot of the prepared solution (see 4.3.1) and dilute with distilled water to about 150 ml. Add 10 ml. of concentrated hydrochloric acid and a few drops of nitric acid; heat to boiling. Add 3 drops of methyl red indicator and a little ashless filter paper pulp, neutralize with concentrated ammonium hydroxide, add 3 drops in excess, and boil for 2 minutes. Filter, after allowing the precipitate to settle, through a 9 cm. Whatman No. 41 (or equivalent) filter paper. Return the filter paper and contents to the same beaker, add 10 ml. of concentrated hydrochloric acid and macerate the paper to a pulp with the aid of the stirring rod. Add 150 ml. of hot distilled water and 3 drops of methyl red indicator and heat to boiling. Add an excess of 3 drops of concentrated ammonium hydroxide and boil for 2 minutes. Filter through an 11 cm. Whatman No. 41 (or equivalent) paper, wash 6 times with hot 2 percent ammonium chloride solution and ignite the precipitate to constant weight at 1,200° C.

$$\text{Percent aluminum oxide } (Al_2O_3) = \frac{100\ A}{W}$$

where:

A = weight of residue
W = weight of sample in aliquot.

4.3.4 *Ammonia content.*

4.3.4.1 *Apparatus.*—Set up a distillation apparatus with the following; long-necked pyrex flask (Kejeldahl flask); spray trap attached at the top of flask; glass condenser; and adapter tube at end of condenser dipping into a 250 ml. Erlenmeyer flask containing

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the standard acid solution. Care must be used in setting up the apparatus since small leaks will cause poor results.

4.3.4.2 *Procedure.* — Dissolve 2 gm. of sample alum in 150 ml. of distilled water in a distillation flask. Pipette 50 ml. of 0.1 *N* sulfuric acid into receiver flask. (Tip of adapter should be below acid surface.) Pour 20 ml. excess of 50 percent sodium hydroxide into the distillation flask, add 100 ml. distilled water, quickly connect to the trap and condenser, and shake contents of flask with a rotary motion. Heat the flask immediately and rapidly bring to a boil. Careful attention must be given to the equipment during distillation. When distillation is carefully performed there should be a gentle oscillation of a short column of the acid in the lower end of the adapter, but no escape of bubbles after air has been expelled. Distillation equilibrium must be maintained or the contents of the receiver will suck back into the distillation flask. Collect about 100 ml. of distillate. Titrate excess of standard acid with 0.1 *N* sodium hydroxide.

$$\text{Percent ammonia } (NH_3) = \frac{1.703\ (AN\text{-}BM)}{W}$$

where:

A = ml. sulfuric acid used in absorption of NH_3
N = normality of sulfuric acid
B = ml. sodium hydroxide used in back titration
M = normality of sodium hydroxide
W = weight of sample.

4.3.5 *Sulfur trioxide content.* — Take a 100-ml. aliquot of the prepared solution (see 4.3.1). Add 5 ml. reagent-grade hydrochloric acid. Dilute solution to 250 ml. with distilled water. Add slowly 20 ml. of 10-percent barium chloride solution and allow to stand overnight or for a minimum of 3 hours. Filter through a tared crucible. Wash thoroughly, ignite, cool, and weigh.

$$\text{Percent sulfur trioxide } (SO_3) = \frac{34.3A}{W}$$

where:

A = weight of barium sulfate
W = weight of sample in aliquot.

4.3.6 *Particle size.* — Sieve tests shall be made in accordance with the applicable sections of Federal Specification P–S–536.

5. PREPARATION FOR DELIVERY

5.1 Material for immediate use.

5.1.1 *Packaging.*—Packaging shall be such that it will protect and assure the delivery of the commodities in suitable condition to the initial Government agency that will use the item.

5.1.2 *Packing.*—Packing shall be adequate to insure delivery of the commodity in a useable condition at the lowest applicable transportation rate. Packing complying with the rules of the published rate classifications, trucking associations, postal or Railway Express agencies will be considered adequate.

5.1.3 *Marking.*—Marking shall be in accordance with 5.2.4.

5.2 Packaging and packing for domestic shipment and storage or overseas shipment.

5.2.1 *Unit packaging.* — Unless otherwise specified, alum in the unit quantities specified in the contract or order shall be packaged as indicated below. The capacities of the unit containers shall be sufficient to contain the quantities specified plus the required outage without waste of space.

Class 1, nut sizes.—Shall be packaged in 1- or 5-pound cans, 85- or 100-pound fiberboard containers, or 85- or 100-pound plywood containers, conforming to the type IV requirements of Military Specification MIL–C–124, and the requirements of Military Specifications JAN–D–111 and JAN–P–112, respectively.

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Class 2, capsule.—80 capsules in 8-ounce wide-mouth square bottle labeled "Alum".

5.2.1.1 *Outage.*—Outage, variously termed "vacant space", "voids", or "head space" required in the unit container depends upon the coefficient of expansion of the contents and the change in temperature to which it may be subjected in transit and storage. Outage should be calculated to the total capacity of the container. Sufficient outage shall be provided in unit containers to prevent leakage or distortion of containers due to the expansion of contents from change of temperature during transit or storage.

5.2.1.2 *Closure.*—All containers shall be securely closed in accordance with standard practice for the type container, unless otherwise specified in the applicable container specification including the appendices thereto.

5.2.2 *Intermediate packaging.*

5.2.2.1 *Bottles, plastic containers, and cans.*—Except as otherwise specified hereinafter unit packages of the same chemical in containers of the same type and size shall be packaged with closures on top in snug-fitting paperboard set-up boxes conforming to Military Specification JAN–P–133, and fiberboard boxes conforming to Federal Specifications LLL–B–631, LLL–B–636, and JAN–P–108, respectively, compliance symbol as applicable, in quantities and arrangement as specified in table I. Intermediate packaging is not required for chemicals in cans of more than 5-pound capacity. If the quantities specified are by volume they shall be converted to the corresponding weights in applying table I. Where such weights are not listed in table I the next higher weight listed shall be used.

5.2.2.2 *Cushioning.*—Fiberboard boxes shall have full-height partitions, vertical corrugated liners, and full-size top and bottom pads. Separators shall be placed between all layers of unit packages. The cells formed by the partitions shall be of such size that the unit containers contact all 4 sides of the cell, and if necessary, additional cushioning shall be effected by insertion of cellulose wadding or similar nonhygroscopic material satisfactory to the Contracting Officer, to prevent movement within the boxes. Paperboard set-up boxes shall have slotted partitions of paperboard and additional cushioning of cellulose wadding when necessary to make a tight pack. Closures or handles of cans extending above the tops of the cans shall be protected by fiberboard pads. The dimensions and arrangement of the pads shall be such that any bearing on the closure or handles will be prevented.

5.2.2.3 *Closures.*—Fiberboard boxes shall be closed and sealed in conformance with the requirements of the appendix to the box specification, and in addition by covering the full length of all joints and seams, including manufacturer's joints, with gummed tape of 3-inch minimum width, extending tape over corners and edges at least 3 inches. Paperboard set-up boxes shall have covers secured with gummed tape of suitable width.

5.2.2.4 *Drums, or fiberboard boxes.*—Unless otherwise specified, intermediate packaging is not required for chemicals in drums, or fiberboard boxes.

TABLE II.—*Intermediate packaging of bottles or cans*

Unit weight of contents	Arrangement of unit packages in fiberboard boxes			Total number or unit packages
	Number of layers	Number of rows	Number in row	
8 ounces	2	4	4	32
1 pound	1	4	4	16
5 pounds	1	1	2	2
over 5 pounds (except cans)	1	1	1	1

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5.2.3 *Packing.*

5.2.3.1 *For domestic shipment and storage.*—Unless otherwise specified the subject commodity packaged as described in 5.2 shall be packed in wood-cleated fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood or wood-cleated boxes shall not exceed approximately 150 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification. Drums will require no overpacking.

5.2.3.2 *For overseas shipment.* — When specified the subject commodity packaged as described in 5.2 shall be packed in wood-cleated solid fiberboard, cleated-plywood, nailed wood, wirebound, fiberboard, or fiber-lined wood boxes, conforming to Military Specifications JAN–P–103, JAN–P–105, JAN–P–106, MIL–B–107, JAN–P–108, and JAN–P–138, respectively. The gross weight of wood or wood cleated boxes shall not exceed approximately 100 pounds, and of fiberboard boxes approximately 70 pounds. The containers shall be fitted with a waterproof case liner conforming to the type I, grade B, class 2 requirements of Military Specification MIL–L–10547. Seams and closures shall be sealed with adhesive conforming to Military Specification MIL–A–140. Drums will require no overpacking.

5.2.4 *Marking.*

5.2.4.1 *Label.* — Each bottle of capsules shall be labeled as follows:

Alum, Ammonia

WARNING: This material is ammonia alum, 1 part ammonia alum equals ⅔ part aluminum sulfate (filter alum).

Each can or drum shall be labeled with the name and class of alum, the quantity contained therein, and the name of the manufacturer or contractor.

5.2.4.2 *Packaging.* — Unless otherwise specified, packages shall be marked with the name and class of alum, the quantity contained therein and the name of the manufacturer or contractor.

5.2.4.3 *Shipping containers.* — Unless otherwise specified, shipping containers shall be marked with the name and class of the commodity as defined in the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, the number of the contract, the name and address of the consignee and the gross weight.

5.2.4.4 *Marking for all Department of Defense orders.* — Interior packages (unit or intermediate) as applicable and all shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** — Purchase documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class required (see 1.2).

These documents should also specify whether the subject commodity is to be packed for immediate use, domestic, or overseas shipment (see 5.1, 5.2, and 5.3).

6.2 Invitations for bids and contracts or orders should require that the contractor furnish a certificate that the alum shall meet requirements of 3.2 and 3.3.

6.3 **Transportation description.**

6.3.1 *Transportation description applicable to these items is:*

Alum, not otherwise indexed by name.
Carload weight 40,000 pounds.
Truckload volume weight 40,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United

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